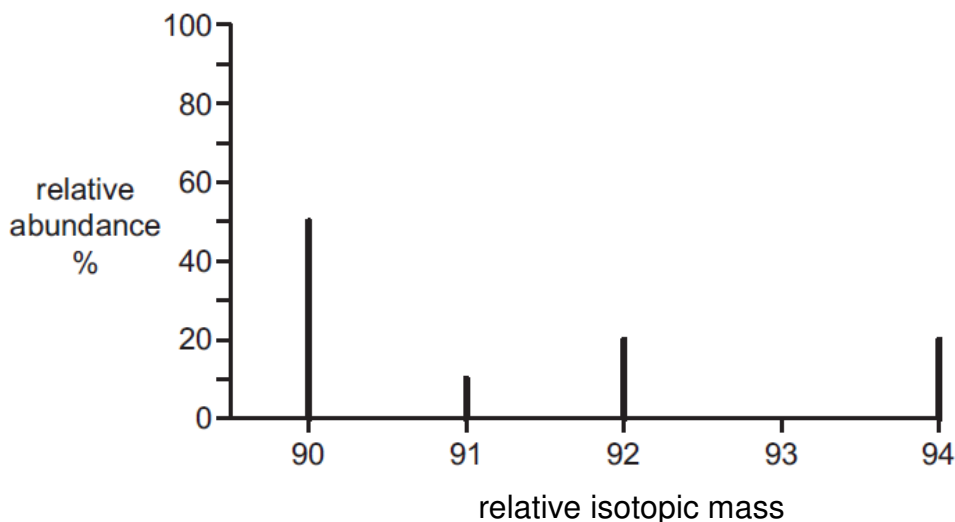


Section A

For each question there are four possible answers, **A**, **B**, **C** or **D**. Choose the **one** you consider to be correct.

- 1 An element **P** consists of four isotopes. The diagram shows the relative abundance of these four isotopes.



What is the relative atomic mass of **P**?

- A** 91.00 **B** 91.30 **C** 91.75 **D** 92.00

- 2 *Use of the Data Booklet is relevant to this question.*

Nickel makes up 20% of the total mass of a coin. The coin has a mass of 10.0 g.

How many nickel atoms are in the coin?

- A** 2.05×10^{22} **B** 4.30×10^{22}
C 1.03×10^{23} **D** 1.20×10^{24}

- 3 A pure hydrocarbon is used in bottled gas for cooking and heating.

When 10 cm³ of the hydrocarbon is burned in 70 cm³ of oxygen (an excess), the final gaseous mixture contains 30 cm³ of carbon dioxide and 20 cm³ of unreacted oxygen. All gaseous volumes were measured under identical conditions.

What is the formula of the hydrocarbon?

- A** C₂H₆ **B** C₃H₆ **C** C₃H₈ **D** C₄H₁₀

- 4 Sulfur dioxide, SO_2 , is added to wines to prevent oxidation of ethanol by air. To determine the amount of SO_2 , a sample of wine is titrated with iodine, I_2 . In this reaction, one mole of SO_2 is oxidised by one mole of I_2 .

What is the change in oxidation number of sulfur in this reaction?

- A** +2 to +4 **B** +2 to +6 **C** +4 to +5 **D** +4 to +6

- 5 The elements radon (Rn), francium (Fr) and radium (Ra) have consecutive proton numbers in the Periodic Table.

What is the order of their first ionisation energies?

	least endothermic	$\longrightarrow$	most endothermic
A	Fr	Ra	Rn
B	Fr	Rn	Ra
C	Ra	Fr	Rn
D	Rn	Ra	Fr

- 6 The table gives the successive ionisation energies for an element **Q**.

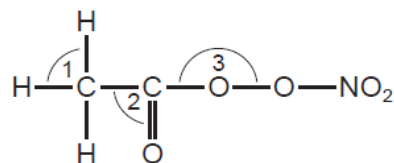
	1st	2nd	3rd	4th	5th	6th
ionisation energy / kJ mol^{-1}	950	1800	2700	4800	6000	12300

What could be the formula of the chloride of **Q**?

- A** QCl **B** QCl_2 **C** QCl_3 **D** QCl_4

- 7 Organic nitrates in photochemical smog can cause breathing difficulties.

The diagram shows an example of an organic nitrate molecule.

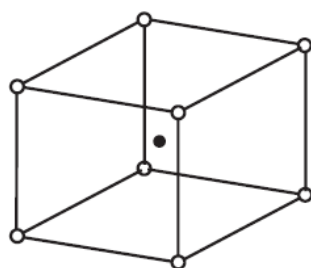


What is the correct order of the bond angles shown in ascending order (smallest first)?

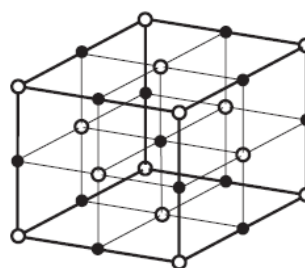
- A** $3 \rightarrow 2 \rightarrow 1$ **B** $3 \rightarrow 1 \rightarrow 2$ **C** $2 \rightarrow 1 \rightarrow 3$ **D** $1 \rightarrow 2 \rightarrow 3$
- 8 The table below gives the ionic radii, in pm, of some ions. [$1 \text{ pm} = 10^{-12} \text{ m}$]

ion	ionic radius / pm
Na^+	102
Mg^{2+}	72
Cs^+	167
Cl^-	181
O^{2-}	140

Caesium chloride, CsCl , has a different lattice structure from both sodium chloride, NaCl , and magnesium oxide, MgO .



CsCl lattice



NaCl and MgO lattice

Which factor appears to determine the type of lattice for these three compounds?

- A** the charge on the cation **B** the ratio of the ionic radii
C the ratio of the ionic charges **D** the sum of the ionic charges

- 9 Hydrogen bonding can occur between molecules of methanal, HCHO , and molecules of liquid **R**.

What could liquid **R** be?

- | | |
|-------------------------------------|--|
| A CH_3OH | B CH_3CHO |
| C CH_3COCH_3 | D $\text{CH}_3\text{CO}_2\text{CH}_3$ |

- 10 Given the following enthalpy changes:



what is the enthalpy change of reaction, $\Delta H^\ominus$, of $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$?

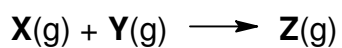
- | | |
|-------------------------------------|-------------------------------------|
| A -196 kJ mol^{-1} | B -98 kJ mol^{-1} |
| C $+98 \text{ kJ mol}^{-1}$ | D $+196 \text{ kJ mol}^{-1}$ |

- 11 In a calorimetric experiment, 1.60 g of a fuel is burnt. 45% of the energy released is absorbed by 200 g of water whose temperature rose from 18°C to 66°C . The specific heat capacity of water is $4.20 \text{ J g}^{-1} \text{ K}^{-1}$.

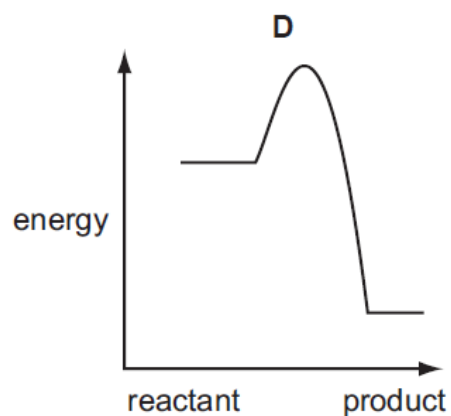
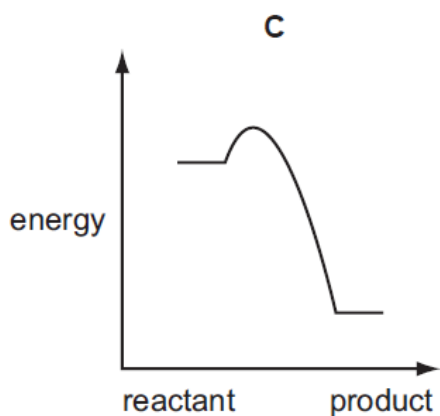
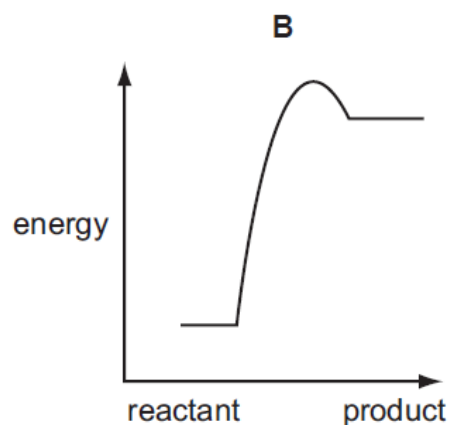
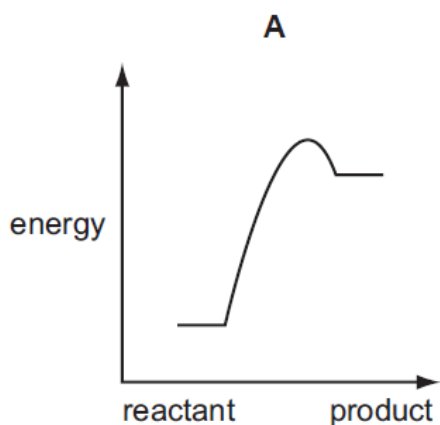
What is the total energy released per gram of fuel burnt?

- | | |
|-------------------|--------------------|
| A 25 200 J | B 56 000 J |
| C 89 600 J | D 144 000 J |

12 Four reactions of the type shown below, are studied at the same temperature:



Which is the correct reaction pathway diagram for the reaction that would proceed most rapidly and with good yield?



- 13 Use of the Data Booklet is relevant to this question.

Water dissociates according to the equation below:

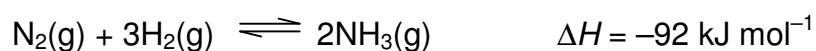


At 25 °C, the equilibrium value of $[\text{H}^+]$ is $10^{-7} \text{ mol dm}^{-3}$ and $[\text{H}_2\text{O}] = \frac{1000}{18} \text{ mol dm}^{-3}$.

Which of the following lists the numerical values of pH, $\text{p}K_{\text{a}}$ and $\text{p}K_{\text{w}}$ of water in the correct ascending order?

	<i>smallest</i>		<i>largest</i>
A	pH	$\text{p}K_{\text{a}}$	$\text{p}K_{\text{w}}$
B	pH	$\text{p}K_{\text{w}}$	$\text{p}K_{\text{a}}$
C	$\text{p}K_{\text{a}}$	$\text{p}K_{\text{w}}$	pH
D	$\text{p}K_{\text{w}}$	$\text{p}K_{\text{a}}$	pH

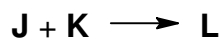
- 14 The Haber process for the manufacture of ammonia is represented by the following equation.



Which statement is correct about this reaction when the temperature is increased?

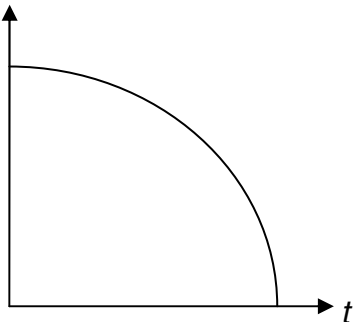
- A** There is no effect on the backward or forward rate.
- B** The backward rate only increases.
- C** The forward rate only increases.
- D** Both forward and backward rates increase.

- 15 The product of the concentrations of **J** and **K**, $[J][K]$, is plotted against time, t , for the following second order reaction:

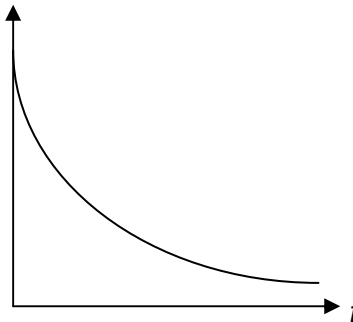


Which graph would be obtained?

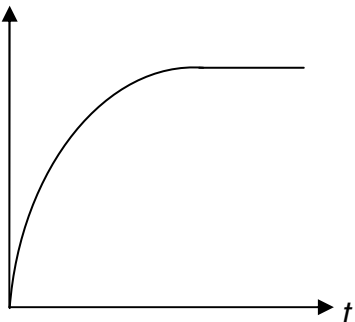
A $[J][K]$



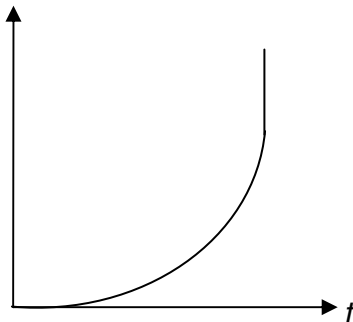
B $[J][K]$



C $[J][K]$



D $[J][K]$



- 16 When dangerous chemicals are transported on the road, the vehicles must carry signs that indicate the measures that should be taken in the event of a chemical spillage.

Which material must be used if metallic sodium was spilled?

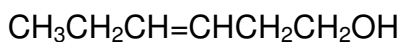
A ethanol

B acidic spray

C sand

D jets of water

- 17 The compound 'leaf alcohol' is partly responsible for the smell of new-mown grass.

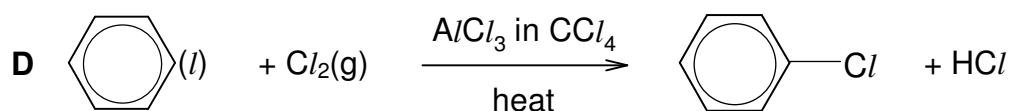
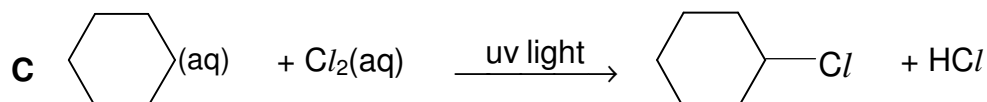
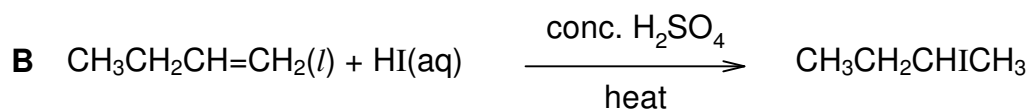
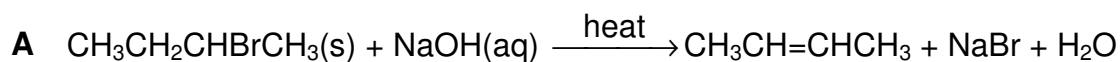


leaf alcohol

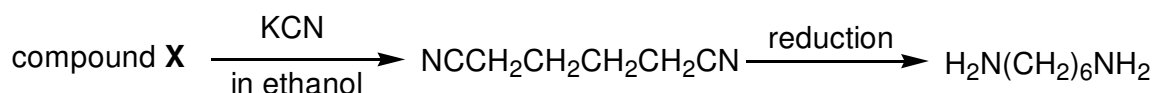
Which two compounds will be formed when 'leaf alcohol' is oxidised using hot, concentrated manganate(VII) ions?

- A $\text{CH}_3\text{CO}_2\text{H}$ and $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 B $\text{CH}_3\text{CO}_2\text{H}$ and $\text{HO}_2\text{CCH}_2\text{CH}_2\text{CO}_2\text{H}$
 C $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ and $\text{HOCH}_2\text{CH}_2\text{CO}_2\text{H}$
 D $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ and $\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$
- 18 The reaction conditions for four different transformations are given.

Which transformation has the correct conditions?



- 19 The reaction scheme outlines the production of one of the monomers of nylon-6,6 from compound **X**:



Which compound could be **X**?

- A $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
 - B $\text{CH}_2=\text{CHCH}=\text{CH}_2$
 - C $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 - D $\text{HO}_2\text{CCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
- 20 Four drops of 1-chlorobutane, 1-bromobutane and 1-iodobutane were put separately into three test-tubes containing 1.0 cm^3 of aqueous silver nitrate at 60°C .

A hydrolysis reaction occurred.

(R represents the butane chain C_4H_9- and X the halogen atom.)



The rate of formation of cloudiness in the tubes was in the order $\text{RCI} < \text{RBr} < \text{RI}$. Why is this so?

- A The R-X bond polarity decreases from RCI to RI.
 - B The solubility of AgX decreases from AgCI to AgI.
 - C The ionisation energy of the halogen decreases CI from to I.
 - D The bond energy of R-X decreases from RCI to RI.
- 21 Chlorofluorocarbons, CFCs, have been widely used as refrigerants, aerosol propellants and in making foamed plastics, but are now known to destroy ozone in the upper atmosphere.

What will **not** destroy ozone, and can therefore be used safely as a replacement for CFCs?

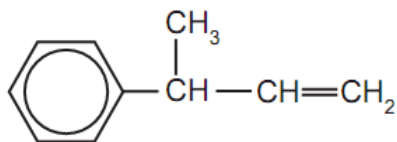
- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- B $\text{CHCl}_3/\text{CFCl}_3/\text{F}_2$
- C CCl_3CBr_3
- D CHBr_3

22 Compound Y

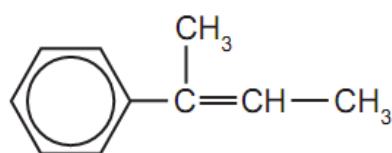
- has the molecular formula $C_{10}H_{14}O$;
- is unreactive towards warm acidified potassium dichromate(VI).

Which of the following could be a compound formed by the dehydration of Y?

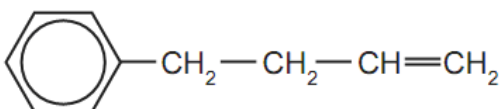
A



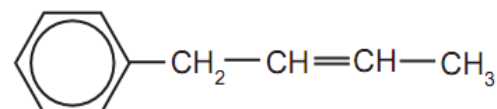
B



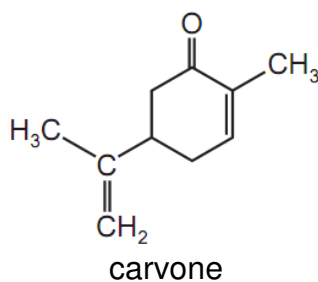
C



D



23 Carvone gives the characteristic flavour to caraway and spearmint.



Prolonged heating of carvone with concentrated acidified potassium manganate(VII) produces carbon dioxide and compound X.

X contains nine carbon atoms and reacts with 2,4-dinitrophenylhydrazine reagent.

What is the maximum number of molecules of 2,4-dinitrophenylhydrazine that will react with one molecule of X?

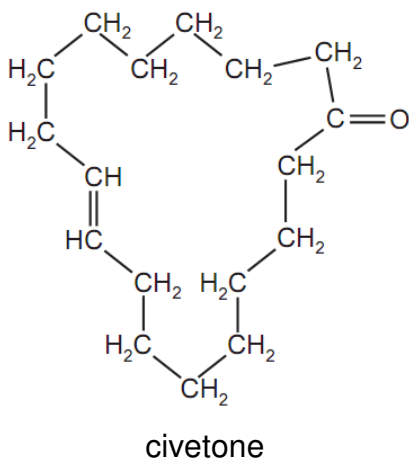
A 1

B 2

C 3

D 4

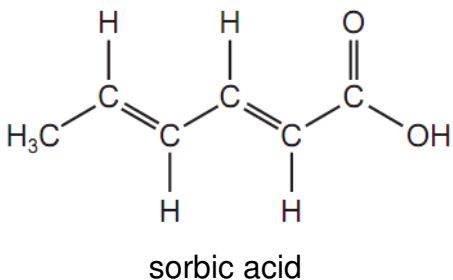
- 24** The naturally-occurring molecule civetone is found in a gland of the African civet cat and has been used in perfumes.



Which reagent does **not** react with civetone?

- A** Fehling's solution
B hydrogen bromide
C 2,4-dinitrophenylhydrazine
D sodium borohydride

- 25** Sorbic acid is used as a food preservative because it kills fungi and moulds.



Sorbic acid will react with

- hydrogen in the presence of a nickel catalyst
- bromine in an organic solvent

How many moles of hydrogen and bromine will be incorporated into one mole of sorbic acid by these reactions?

	moles of hydrogen	moles of bromine
A	2	2
B	2	2.5
C	3	2
D	3	2.5

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

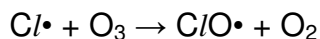
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** The chlorine oxide free radical, $\text{ClO}\cdot$, is formed during the depletion of the ozone layer by chlorofluoroalkanes (CFCs).



Which features are present in the chlorine oxide free radical?

- 1** an odd number of electrons
 - 2** a single covalent bond
 - 3** a dative covalent bond from oxygen to chlorine
- 27** The Group IV elements carbon, silicon and germanium all exist in a diamond structure. The bond lengths in these structures are given below.

element <i>X</i>	C	Si	Ge
bond length <i>X</i> – <i>X</i> / nm	0.154	0.234	0.244

Why does the bond length increase down the group?

- 1** Atomic radius increases down the group.
- 2** Orbital overlap decreases down the group.
- 3** Nuclear charge increases down the group.

The responses **A** to **D** should be selected on the basis

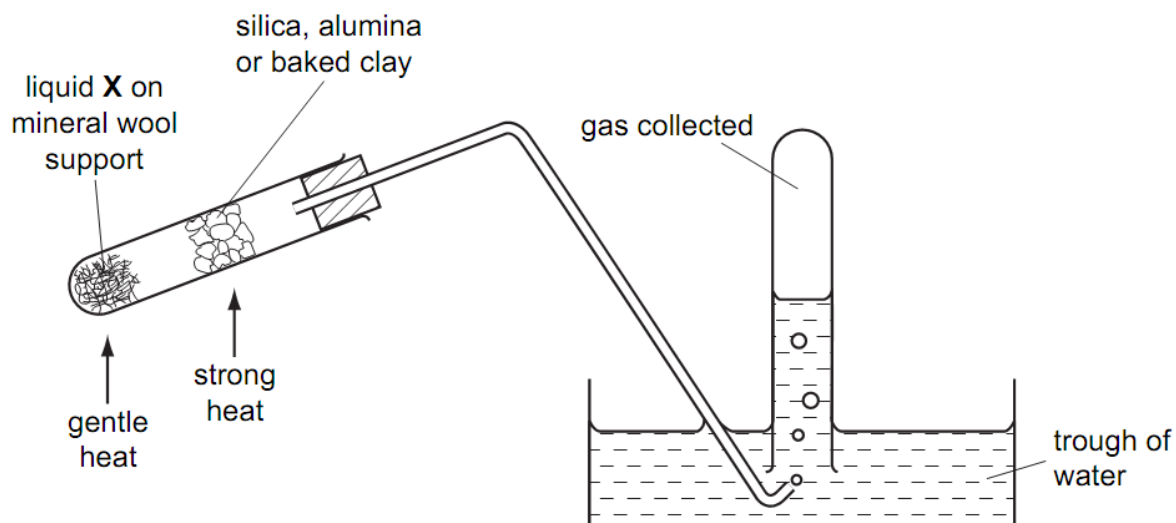
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 In the Periodic Table, the electronegativity of the elements increases

- 1** from sodium to chlorine in Period 3.
- 2** from iodine to fluorine in Group VII.
- 3** from barium to beryllium in Group II.

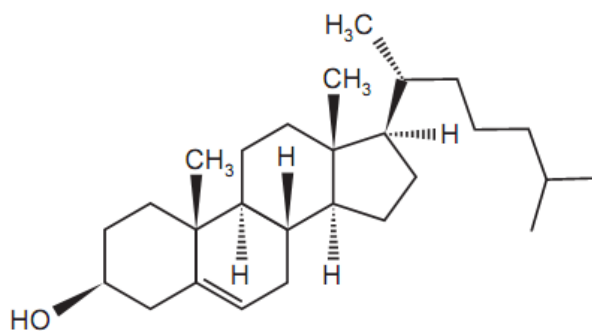
29 The diagram below shows the apparatus used for an experiment.



Which processes could be demonstrated by using the above apparatus?

- 1** the oxidation of ethanol (the liquid **X**)
- 2** the dehydration of ethanol (the liquid **X**)
- 3** the cracking of paraffin (the liquid **X**)

30 The diagram shows the structure of the naturally-occurring molecule cholesterol.



cholesterol

Which statements about cholesterol are correct?

- 1 The molecule contains a secondary alcohol.
- 2 All carbon atoms in the four rings lie in the same plane.
- 3 The molecule contains two π bonds.

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